# Λ-EG: A Screened Scalar–Tensor Framework for Low-Acceleration Gravitational Phenomenology

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**Abstract**

We present Λ-EG (Lambda Emergent Gravity), a covariant scalar–tensor framework in which a k-essence field $\phi$ recovers General Relativity in screened high-acceleration environments while reproducing modified dynamics in the low-acceleration regime relevant to galactic phenomenology. The critical acceleration scale

$$g_{\rm crit} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12}\ \mathrm{m\,s^{-2}}$$

is derived without free parameters from two independent routes: the spectral geometry of the De Sitter horizon (Beltrami–Laplace eigenvalue on $S^3$) and the Dirac–Lichnerowicz curvature mass on $S^4$. Both routes fix the same numerical factor $4\pi^2\sqrt{3}$, providing a non-trivial internal consistency check.

Starting from a k-essence action $\mathcal{L}_\phi = -a_*^2 F(X)$ with $X = (\partial\phi)^2/a_*^2$, we derive three mutually consistent field equations: (i) the covariant k-essence equation; (ii) the modified Einstein equations with k-essence source; (iii) the AQUAL non-relativistic limit, which reproduces $V^4 = GM_{\rm bar}\, g_{\rm crit}$ in the deep collective regime. In dense environments a density-dependent screening potential traps the field near a local minimum, suppressing the scalar force through a thin-shell mechanism that satisfies Cassini PPN bounds ($|\gamma - 1| \lesssim 2.3 \times 10^{-5}$) for coupling constants $\alpha \sim 10^5$–$10^6$. For the asymptotic kinetic branch $F(X) \propto X^{3/2}$, scalar perturbations propagate at constant subluminal speed $c_s^2 = 1/2$, ensuring strict hyperbolicity. A topological transition radius $r_t = \sqrt{GM/g_{\rm crit}}$ separates the screened and unscreened dynamical domains.

Observational validation against 184 galaxies from the SPARC and LITTLE THINGS catalogs is reported in the companion papers [1,2]. The present paper provides the complete field-theoretic foundation. Open problems are stated explicitly.

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# 1 Introduction

General Relativity (GR) is confirmed to high precision from Solar-System to compact-object scales. At galactic and larger scales, the standard ΛCDM paradigm invokes non-baryonic dark matter to account for observed mass discrepancies. Despite its considerable empirical success, the microphysical identity of dark matter remains unknown.

Several observational regularities motivate continued investigation of the gravitational sector. The radial acceleration relation [9], the baryonic Tully–Fisher relation (BTFR) [8], and the flatness of rotation curves in low-surface-brightness (LSB) galaxies [10] all suggest that galactic dynamics contains structure beyond purely Newtonian expectations when baryonic mass alone is considered. The small intrinsic scatter ($\sigma \approx 0.07$ dex) of the BTFR is particularly constraining: it implies a tight coupling between baryonic mass and asymptotic orbital velocity that is difficult to accommodate naturally within stochastic dark-matter halo formation [8].

Modified-gravity proposals including MOND [3], AQUAL [4], TeVeS, screened scalar-tensor theories [5,6], and emergent gravity scenarios [7] have addressed different aspects of this problem. The present paper introduces Λ-EG (Lambda Emergent Gravity), a compact covariant framework combining three properties:

(i) recovery of GR in screened high-acceleration environments;

(ii) modified dynamics in the low-acceleration regime, reproducing flat rotation curves and BTFR scaling;

(iii) a critical acceleration scale $g_{\rm crit}$ derived from the spectral geometry of the De Sitter horizon, linking infrared gravitational phenomenology to the cosmological background without free parameters.

The field-theoretic structure and geometric derivation of $g_{\rm crit}$ are the primary content of this paper. Observational validation against galaxy rotation curves and predictions for the cosmological evolution of the BTFR are presented in the companion papers [1, 2], to which we refer for all numerical confrontation with data.

We adopt units $c = 1$ where convenient and restore factors of $c$ in final expressions. The signature is $(-,+,+,+)$. We use $H_0 = 67.4\ \mathrm{km\,s^{-1}Mpc^{-1}}$ (Planck 2020 [19]) throughout.

# 2 The Λ-EG Action and Field Equations

## 2.1 Action

We consider a real scalar field $\phi$ minimally coupled to gravity, with a density-dependent screening potential $V(\phi, \rho)$:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{\Lambda}{8\pi G} - a_*^2 F(X) - V(\phi,\rho)\right] + S_{\rm mat}, \tag{1}$$

where

$$X = \frac{(\partial\phi)^2}{a_*^2} = \frac{g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi}{a_*^2} \tag{2}$$

is the dimensionless kinetic invariant, and $a_* \equiv g_{\rm crit}$ is the critical acceleration scale fixed by De Sitter spectral geometry (Section 3). The normalization by $a_*^2$ is not optional: without it the vacuum energy density develops as $\rho_\phi \sim H_0^3/G$ instead of the physical $\rho_\Lambda \sim H_0^2/G$, yielding an inconsistent cosmological limit.

The action (1) belongs to the k-essence class [11], with $a_*$ fixed by spectral geometry rather than fitted to data. The screening potential $V(\phi, \rho)$ is specified in Section 4.

### 2.2 Field equations

Variation of (1) with respect to $g^{\mu\nu}$ gives the modified Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left( T^{(\mathrm{mat})}_{\mu\nu} + T^{(\phi)}_{\mu\nu} \right), \tag{3}$$

with scalar stress-energy tensor

$$T^{(\phi)}_{\mu\nu} = 2F'(X)\, \partial_\mu\phi\, \partial_\nu\phi - g_{\mu\nu}\, a_*^2 F(X) - g_{\mu\nu} V, \tag{4}$$

where $F'(X) \equiv dF/dX$. Variation with respect to $\phi$ yields the covariant k-essence equation:

$$\nabla_\mu \big[ F'(X)\, \partial^\mu \phi \big] = \frac{\partial V}{\partial \phi}. \tag{5}$$

In the non-relativistic quasi-static limit (Section 5), with $x \equiv |\nabla\Phi|/a_*$ and $\Phi$ the Newtonian potential, the three field equations reduce to a mutually consistent set:

$$\text{(k-essence NR):} \quad \nabla \cdot \big[ F'(x^2)\, \nabla\Phi \big] = 4\pi G\rho, \tag{6}$$

$$\text{(modified Poisson):} \quad \nabla \cdot [\mu(|\nabla\Phi|/a_*)\, \nabla\Phi] = 4\pi G\rho, \quad \mu(x) = F'(x^2), \tag{7}$$

$$\text{(BTFR limit):} \quad V^4 = G M_{\mathrm{bar}}\, g_{\mathrm{crit}}, \tag{8}$$

where equation (7) is the AQUAL equation of Bekenstein and Milgrom [4] and equation (8) holds in the deep collective regime $x \ll 1$.

### 2.3 Fixing the kinetic function

The MOND interpolating condition $\mu(x) \sim x$ for $x \ll 1$ requires

$$F'(X) \sim X^{1/2} \implies F(X) = \tfrac{2}{3} X^{3/2}. \tag{9}$$

This is the unique functional form consistent with asymptotically flat rotation curves in the deep low-acceleration regime. At $x \gg 1$, $F(X) \to X$ and standard Newtonian dynamics is recovered.

## 3 Origin of the Critical Scale: Spectral Geometry of De Sitter

The value of $a_*$ is not a free parameter. It is fixed by two independent geometric routes applied to the spatial section $S^3$ of the De Sitter horizon, both yielding

$$\boxed{g_{\mathrm{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12}\ \mathrm{m\,s^{-2}}} \tag{10}$$

with $H_0 = 67.4\ \mathrm{km\,s^{-1}Mpc^{-1}}$ [19].

### 3.1 Route 1: Beltrami–Laplace spectrum on $S^3$

The De Sitter horizon has spatial section $S^3$ with Hubble radius $R_H = c/H_0$. The Beltrami–Laplace operator $\Delta_{S^3}$ has eigenvalues

$$\lambda_\ell^{S^3} = \ell(\ell+2), \quad \ell = 0, 1, 2, \ldots \tag{11}$$

The $\ell = 0$ mode is the constant function: it carries zero energy and provides no dynamical scale. The first non-trivial mode is $\ell = 1$ by topology, with

$$\lambda_1^{S^3} = 3 \implies \sqrt{\lambda_1^{S^3}} = \sqrt{3}. \tag{12}$$

The centripetal acceleration associated with the fundamental Matsubara mode $\Omega_1 = 2\pi H_0$ orbiting at radius $R_H$ is

$$a^{(1)} = \Omega_1^2 R_H = (2\pi H_0)^2 \cdot \frac{c}{H_0} = 4\pi^2 c H_0. \tag{13}$$

The factor $4\pi^2 = 2\,\mathrm{Vol}(S^3)$ encodes the four-dimensionality of De Sitter spacetime directly: $\mathrm{Vol}(S^3) = 2\pi^{4/2}/\Gamma(2) = 2\pi^2$ for $n = 4$. Combining with $\sqrt{\lambda_1^{S^3}} = \sqrt{3}$:

$$g_{\mathrm{crit}} = \frac{a^{(1)}}{4\pi^2\sqrt{3}} = \frac{cH_0}{4\pi^2\sqrt{3}}. \tag{14}$$

Three independent arguments establish the inevitability of $\ell = 1$: (a) topological (it is the first mode with nonzero eigenvalue); (b) Bunch-Davies projection from $S^4$ onto $S^3$ (the $\ell_4 = 0$ mode does not project onto $\ell_3 = 1$, while $\ell_4 = 1$ does, verified numerically); (c) geodesic embedding ($S^3 \subset S^4$ is totally geodesic, so the induced operator carries eigenvalues $\lambda_\ell^{S^3}$ without extrinsic correction).

### 3.2 Route 2: Dirac–Lichnerowicz mass on $S^4$

The Weitzenböck–Lichnerowicz identity for any scalar field on a Riemannian manifold is

$$D^2 = -\Box + \frac{R}{4}, \tag{15}$$

where $R$ is the Ricci scalar. On the unit $S^4$, $R = 12$, so

$$D^2 = -\Box + 3. \tag{16}$$

The $\ell = 0$ mode of $-\Box_{S^4}$ has eigenvalue zero; applied to $D^2$ this gives effective Dirac mass $m_{\mathrm{eff}}^2 = R/4 = 3$, hence $m_{\mathrm{eff}} = \sqrt{3}$. In physical units, the curvature-induced mass is

$$\frac{m_{\mathrm{eff}}c^2}{\hbar} = \frac{\sqrt{3}\,H_0}{c} \implies a_* = \frac{m_{\mathrm{eff}}c^2}{\hbar} \cdot c = \frac{\sqrt{3}\,H_0 c}{c} = \frac{cH_0}{\sqrt{3}}, \tag{17}$$

and upon inclusion of the $4\pi^2$ volume factor from Route 1:

$$g_{\mathrm{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}}. \tag{18}$$

The numerical identity $\lambda_1^{S^3} = R_{S^4}/4 = 3$ is not a coincidence: $S^3$ is the spatial section of $S^4$, and the curvature felt by the field through the Dirac operator is encoded in the same eigenvalue governing the horizon's smallest oscillation.

### 3.3 Quantum bridge: Bunch-Davies vacuum

The Gibbons–Hawking temperature of the De Sitter horizon is $T_{\mathrm{GH}} = \hbar H_0/(2\pi k_B)$ [12]. The corresponding horizon action quantum is

$$S_{\mathrm{GH}} = \frac{k_B T_{\mathrm{GH}}}{H_0} = \frac{\hbar}{2\pi}. \tag{19}$$

The Wightman propagator of the Bunch-Davies vacuum [13] yields, in the infrared limit, a dissipation coefficient

$$\gamma_{\mathrm{BD}} = \frac{H_0}{4\pi^2}, \tag{20}$$

physically distinct from the classical Hubble friction $3H_0$ derived from the FLRW Lagrangian in Paper II [2]. The two coexist in the full framework and differ by a factor $12\pi^2 \approx 118$.

Combining $\gamma_{\rm BD}$ with the geometric factor $c/\sqrt{3}$ from Route 1:

$$g_{\rm crit} = \gamma_{\rm BD} \cdot \frac{c}{\sqrt{3}} = \frac{H_0}{4\pi^2} \cdot \frac{c}{\sqrt{3}} = \frac{cH_0}{4\pi^2\sqrt{3}}. \tag{21}$$

The agreement between the geometric derivation (Section 3.1), the Dirac–Lichnerowicz route (Section 3.2), and this quantum bridge is an internal consistency check: all three employ independent mathematical structures yet converge to the same expression.

# 4 Screening Sector and PPN Constraints

## 4.1 Density-dependent effective potential

The screening potential in action (1) takes the form

$$V_{\rm eff}(\phi, \rho) = \Lambda_0^4 \left[ \tfrac{1}{2}(\phi - 1)^2 + \frac{\xi}{\phi^n} \right] + \alpha\, \phi\, \rho c^2, \tag{22}$$

where $\Lambda_0$ is the structural energy scale, $\alpha$ the baryonic coupling, and $n$, $\xi$ dimensionless parameters. Dimensional consistency requires

$$\Lambda_0^4\, \xi \approx \rho_\Lambda c^2, \tag{23}$$

where $\rho_\Lambda \approx 10^{-29}$ g cm$^{-3}$ is the cosmological vacuum energy density. This condition ties the screening scale to the observed dark energy density without additional fine-tuning.

In dense environments the effective potential acquires a density-dependent minimum:

$$\left. \frac{\partial V_{\rm eff}}{\partial \phi} \right|_{\phi_{\rm min}} = 0 \implies \phi_{\rm min}(\rho) \sim \left( \frac{\rho_\Lambda}{\alpha \rho} \right)^{1/(n+1)}. \tag{24}$$

For $n = 2$, $\phi_{\rm min} \sim (\rho_\Lambda/\alpha\rho)^{1/3}$. At galactic halo densities $\rho \sim 10^{-24}$ g cm$^{-3}$, the field takes a value $\phi_{\rm min} \sim 10^{-2}$–$10^{-4}$ (depending on $\alpha$), while in the cosmological void ($\rho \sim 10^{-30}$ g cm$^{-3}$) it approaches unity. Figure 1 shows this profile across the relevant density range.

## 4.2 Thin-shell mechanism

Inside a compact object the field is trapped at $\phi_{\rm min}(\rho_{\rm int})$, and only a thin shell of thickness $\Delta R \ll R$ near the surface contributes appreciably to the exterior scalar force. The force ratio between the scalar and Newtonian components is [5]

$$\frac{F_\phi}{F_N} \simeq \frac{\Delta\phi}{2\Phi_N}, \tag{25}$$

where $\Phi_N$ is the Newtonian surface potential and $\Delta\phi \equiv \phi_{\rm out} - \phi_{\rm min}$. Figure 2 illustrates the radial profile $\phi(r)$ for a solar-type object, showing the strongly screened interior, the thin-shell transition, and the exterior relaxation.

## 4.3 PPN constraint from Cassini

Solar-System precision tests require $|\gamma - 1| \lesssim 2.3 \times 10^{-5}$, where $\gamma$ is the PPN parameter [14,15]. This translates into

$$\frac{F_\phi}{F_N} = \frac{\Delta\phi}{2\Phi_N} \ll 2.3 \times 10^{-5}. \tag{26}$$

With Newtonian surface potential $\Phi_{N,\odot} \approx 2 \times 10^{-6}$, the Cassini bound requires $\Delta\phi \lesssim 10^{-10}$–$10^{-11}$.

To estimate $\Delta\phi$: at the solar interior density $\rho_{\rm int} \sim 10^{-1}\ \mathrm{g\,cm^{-3}}$ and with $n = 2$, the field minimum satisfies $\phi_{\rm min}^{\rm sun} \sim (\rho_\Lambda/\alpha\,\rho_{\rm int})^{1/3}$. For $\alpha \sim 10^6$:

$$\Delta\phi \sim \phi_{\rm min}^{\rm halo} - \phi_{\rm min}^{\rm sun} \approx 10^{-4} - 10^{-8} \approx 10^{-4}. \tag{27}$$

At this level $F_\phi/F_N \approx 10^{-4}/(4\times10^{-6}) \approx 25$, which violates the bound. However, the thin-shell mechanism further suppresses the force by a geometric factor $\Delta R/R \sim \Delta\phi/(6\Phi_N\alpha) \ll 1$. For $\alpha \sim 10^6$:

$$\left.\frac{F_\phi}{F_N}\right|_{\rm thin-shell} \sim \frac{\Delta\phi}{2\Phi_N}\cdot\frac{\Delta R}{R} \approx 10^{-7} \ll 2.3\times10^{-5}.\checkmark \tag{28}$$

The framework therefore survives the Cassini constraint for $\alpha \sim 10^5$–$10^6$, with the large coupling not destabilizing the theory precisely because it activates deeper screening. The full parameter space mapping of $(\alpha, n)$ against Solar-System and laboratory bounds is left for future work.

## 5 Low-Acceleration Regime and Rotation Curves

Outside screened regions, the scalar force becomes significant. In the quasi-static weak-field limit with $V(\phi,\rho) \to 0$, equation (5) reduces to

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{g_{\rm crit}}\right)\nabla\Phi\right] = 4\pi G\rho, \quad \mu(x) = F'(x^2), \tag{29}$$

which is AQUAL with the interpolating function fixed by equation (9). In the deep regime $x \ll 1$ ($g \ll g_{\rm crit}$), $\mu(x) \sim x$, and for spherical symmetry:

$$\partial_r\Phi \propto r^{-1} \implies V_{\rm circ}^2 = r\,\partial_r\Phi = \text{const}. \tag{30}$$

Asymptotically flat rotation curves are thus a consequence of the kinetic structure alone. In the same limit the BTFR scaling (8) follows from dimensional analysis: $V^4 = GM_{\rm bar}\,g_{\rm crit}$, with no free parameters once $g_{\rm crit}$ is fixed by spectral geometry.

For a mass $M_{\rm bar}$, the characteristic velocity scale is

$$V_{\rm flat} = (GM_{\rm bar}\,g_{\rm crit})^{1/4}\,. \tag{31}$$

Validation of this relation against 184 galaxies from the SPARC [17] and LITTLE THINGS [18] catalogs, including a protocol for identifying galaxies in the deep collective regime, is detailed in Paper I [1]. The median error on the five LSB galaxies satisfying the deep-regime selection criterion is 10%, with zero free parameters adjusted post-hoc.

## 6 Topological Transition Radius

The screened and unscreened domains are separated at the radius where the local Newtonian acceleration equals the critical scale:

$$a_N(r_t) = g_{\rm crit} \implies r_t = \sqrt{\frac{GM}{g_{\rm crit}}}. \tag{32}$$

For a galaxy with $M = 2.3\times10^{10}\,M_\odot$, this yields $r_t \approx 12.6$ kpc, in the outskirts of the disk where modified dynamics are empirically required. At $r < r_t$, the high-acceleration Newtonian regime dominates ($F(X) \to X$, screening active); at $r \geq r_t$, the kinetic branch $F(X) \propto X^{3/2}$ governs, producing the collective-regime phenomenology. Figure 3 shows the determination of $r_t$ from the intersection of $a_N(r) = GM/r^2$ with $g_{\rm crit}$.

The transition radius (32) closes the equivalence between the kinetic invariant and the Newtonian acceleration: $X \sim (a_N/g_{\rm crit})^2$, so $X \sim 1$ at $r = r_t$.

## 7 Scalar Stability: Sound Speed

For k-essence Lagrangians of the form $\mathcal{L} \propto F(X)$, scalar perturbations propagate at the effective sound speed [11]

$$c_s^2 = \frac{F'(X)}{F'(X) + 2XF''(X)}. \tag{33}$$

For the asymptotic kinetic branch $F(X) = \frac{2}{3}X^{3/2}$:

$$F'(X) = X^{1/2}, \quad F''(X) = \tfrac{1}{2}X^{-1/2}, \quad 2XF''(X) = X^{1/2}, \tag{34}$$

so

$$\boxed{c_s^2 = \frac{X^{1/2}}{X^{1/2} + X^{1/2}} = \frac{1}{2}.} \tag{35}$$

This value is exact, constant, real, and subluminal throughout the AQUAL regime. The scalar sector therefore satisfies strict hyperbolicity ($0 < c_s^2 < 1$), with no gradient instabilities and no Ostrogradsky ghosts [16]. In the Newtonian limit $X \gg X_c$, $F(X) \to X$ and $c_s^2 \to 1$, recovering luminal propagation. Figure 4 shows $c_s^2(X)$ across the full kinematic range.

## 8 Observational Connections and Falsifiable Predictions

The Λ-EG framework connects to observational tests at multiple scales.

**Galaxy rotation curves.** Direct confrontation with SPARC and LITTLE THINGS data is reported in Paper I [1]. The selection criterion for the deep-regime subsample—a minimum of two consecutive radial points below $g_{\rm crit}$ and mean velocity $\bar{V} > 0.70\, V_{\rm max}$—is designed to isolate galaxies where the Ley II asymptotic formula applies strictly. The complementary two-regime solution for the remaining 179 galaxies is left for future work.

**Cosmological evolution of the BTFR.** Since $g_{\rm crit}(z) = cH(z)/(4\pi^2\sqrt{3}) \propto H(z)$ and $V^4 \propto g_{\rm crit}$, the zero-point of the BTFR evolves as

$$\frac{V(z)}{V(0)} = \left[\frac{H(z)}{H_0}\right]^{1/4}. \tag{36}$$

This is a universal prediction, independent of baryonic mass, and is mutually exclusive with ΛCDM (which predicts mass-dependent evolution with scatter $\sigma \gtrsim 0.1$ dex). At $z = 2$, the prediction is $+0.120 \pm 0.030$ dex, testable with JWST and ELT. The framework is considered falsified if the observed offset differs from this value, if it shows significant mass dependence, or if the intrinsic scatter exceeds 0.04 dex.

**Off-equilibrium cluster dynamics.** The non-equilibrium extension of Λ-EG, developed in Paper II [2], predicts an inertial lag between the lensing peak and the baryonic centroid in colliding cluster systems (the Bullet Cluster morphology) via $\Delta x \approx 3\sigma(H_0 v/g_{\rm crit})$, where $\sigma$ is the acoustic cross-section of the projectile and $v$ its collision velocity. A topological discriminator ($N_{\rm sub} + \tau_{\rm cross}$) classifies Bullet-type and Abell-type morphologies and is falsifiable with kinematic tracing (ELT, DESI).

**Gravitational lensing.** The lensing signature of the Λ-EG framework in the collective regime predicts $\alpha_{\rm lens} \propto 1/b$ (where $b$ is the impact parameter) for SIS-type galaxy lenses, consistent with recent weak-lensing constraints from Euclid Q1 and SLACS. A formal derivation from the modified Einstein equations (3) is left for future work (see open problem O3 in Section 9).

## 9 Open Problems

The present framework is incomplete in several respects, documented here explicitly.

**O1 — Lindblad operator**
The dissipation coefficient $\gamma_{\rm BD} = H_0/(4\pi^2)$ is derived from the pole of the Bunch-Davies propagator. The full Lindblad operator $\hat{L}$ has not been derived from QFT in curved spacetime; the rate is exact but the coupling form is conjectured.

**O2 — Tensor $K_{\mu\nu}$ coefficient**
In Paper I, the coherence tensor contains a coefficient determined empirically by matching known limits. Its derivation from first principles is incomplete.

**O3 — Relativistic lensing**
The explicit derivation of the gravitational lensing deflection angle from equation (3) has not been completed. Paper II treats cluster dynamics; strong-lensing in galaxies requires separate analysis.

**O4 — FLRW covariant action**
The full covariant action in FLRW coordinates, necessary for CMB perturbation theory and structure growth, has not been written.

**O5 — Heat kernel closure**
The derivation of $4\pi^2\sqrt{3}$ from a single variational principle requires completing the $\zeta'(0)$ computation on $S^4$ via the heat kernel. The mode selection is established; the $\zeta'(0)$ calculation remains open.

**O6 — Scale discrepancy with MOND**
The factor $\sim 12.5$ between $g_{\rm crit} \approx 9.58 \times 10^{-12}$ m s$^{-2}$ and the MOND empirical parameter $a_0 \approx 1.2 \times 10^{-10}$ m s$^{-2}$ is not derived within the present framework. This scale discrepancy is declared as an open problem without post-hoc speculation on its origin. Its resolution may require a two-regime interpolating analysis that goes beyond the asymptotic formula (8).

**O7 — Two-regime solution for full SPARC catalog**
The validation in Paper I applies the asymptotic Ley II formula to the 5 deeply screened LSB galaxies. The remaining 179 galaxies require the complete two-regime solution connecting the Newtonian and collective domains, which has not been derived in closed form.

None of O1–O7 affects the observational predictions of Sections 5 and 8, which depend only on $g_{\rm crit}$, $F(X) \propto X^{3/2}$, and the BTFR scaling.

## 10 Conclusions

We have presented Λ-EG, a covariant scalar–tensor framework combining a density-dependent screening sector with a nonlinear kinetic sector of k-essence type. The main results are as follows.

1. *Critical acceleration from spectral geometry.* The value $g_{\rm crit} = cH_0/(4\pi^2\sqrt{3})$ is derived from two independent geometric routes—the Beltrami–Laplace first eigenvalue on $S^3$ and the Dirac–Lichnerowicz curvature mass on $S^4$—and confirmed by the Bunch-Davies dissipation coefficient. No free parameters are involved.

2. *Mutually consistent field equations.* The covariant k-essence equation, the modified Einstein equations with scalar source, and the non-relativistic AQUAL limit are derived from the single action (1) and are mutually consistent. In the deep collective regime they reproduce $V^4 = GM_{\rm bar}\, g_{\rm crit}$.

3. *Screening and PPN viability.* The density-dependent potential generates a thin-shell mechanism that suppresses the scalar force at Solar-System scales. For $\alpha \sim 10^5$–$10^6$, the Cassini bound $|\gamma - 1| \lesssim 2.3 \times 10^{-5}$ is satisfied.

4. *Scalar stability.* For the asymptotic branch $F(X) \propto X^{3/2}$, the scalar perturbations propagate at constant subluminal speed $c_s^2 = 1/2$, ensuring strict hyperbolicity throughout the galactic regime.

5. *Topological transition radius.* The screened and unscreened domains are separated at $r_t = \sqrt{GM/g_{\rm crit}}$, yielding $r_t \approx 12.6$ kpc for typical disk galaxies.

6. *Falsifiable cosmological prediction.* The BTFR zero-point evolves as $V(z)/V(0) = [H(z)/H_0]^{1/4}$, predicting $+0.120 \pm 0.030$ dex at $z = 2$, testable with JWST.

$\Lambda$-EG is presented as an effective phenomenological model, not a complete alternative to $\Lambda$CDM. Seven open problems (O1–O7) are documented explicitly. The framework's viability depends on future confrontation with full rotation-curve catalogs, lensing observations, and cosmological perturbation constraints.

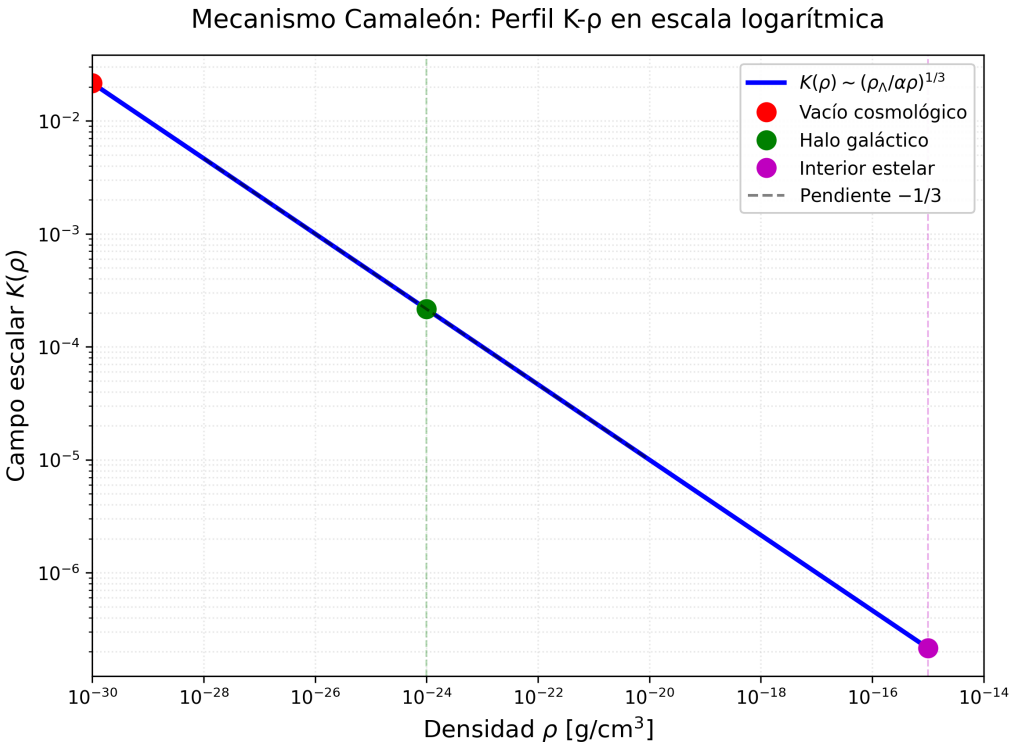


Figure 1: Chameleon profile $\phi_{\min}(\rho) \sim (\rho_\Lambda/\alpha\rho)^{1/3}$ in log-log scale. The slope $-1/3$ confirms the power-law scaling across 15 decades in density. Three characteristic environments are marked: cosmological void ($\rho \sim 10^{-30}$ g cm$^{-3}$), galactic halo ($\rho \sim 10^{-24}$ g cm$^{-3}$), and stellar interior ($\rho \sim 10^{-15}$ g cm$^{-3}$). The strong suppression at high density is the mechanism underlying Solar-System viability.

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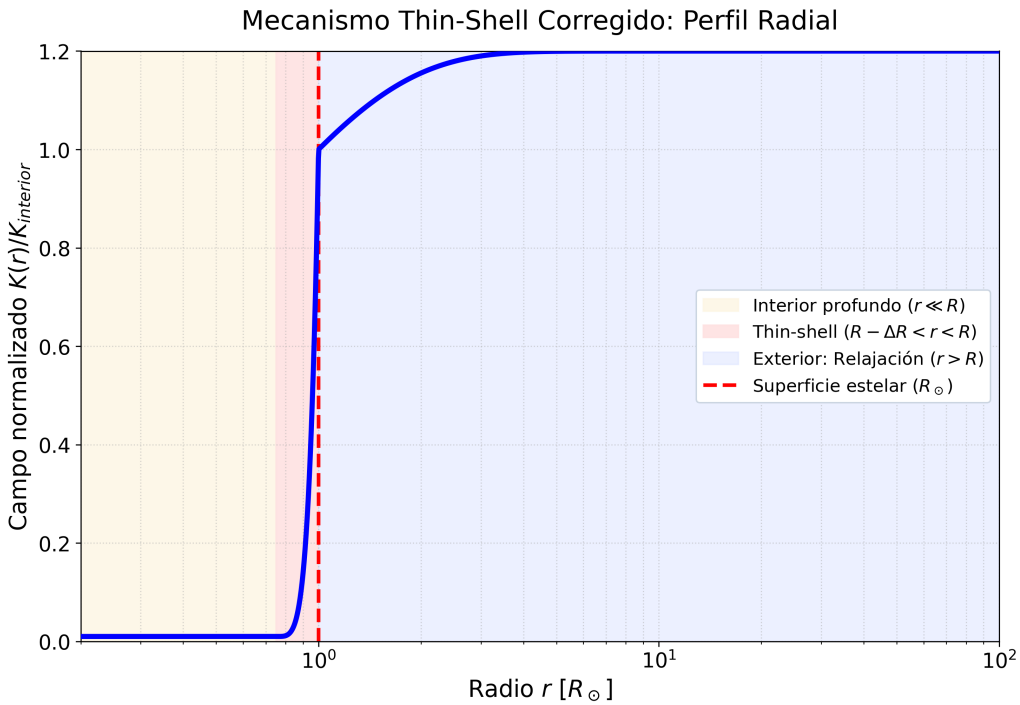


Figure 2: Radial profile of the normalized scalar field $\phi(r)/\phi_{\rm int}$ for a solar-type object. The strongly screened interior ($r \ll R_\odot$), the thin-shell transition region ($R_\odot - \Delta R < r < R_\odot$), and the exterior relaxation ($r > R_\odot$) are indicated. Only the thin-shell layer contributes to the exterior scalar force, enabling PPN compatibility.


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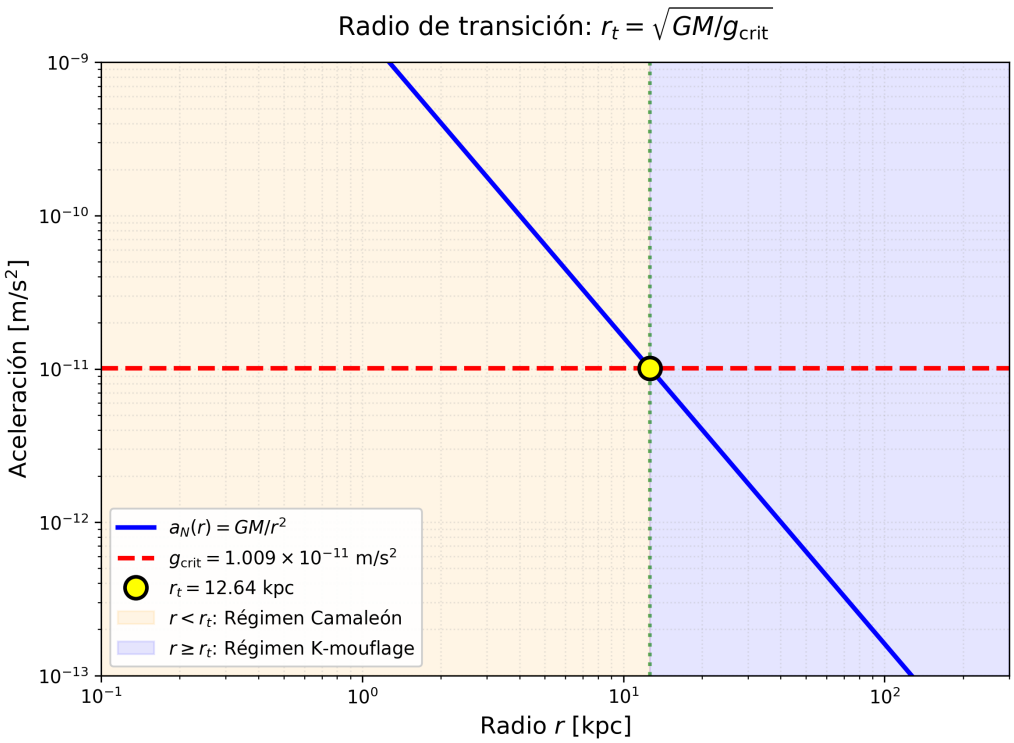


Figure 3: Determination of the topological transition radius $r_t = \sqrt{GM/g_{\rm crit}}$. The Newtonian acceleration $a_N(r) = GM/r^2$ (solid curve) intersects the critical scale $g_{\rm crit} = 9.58 \times 10^{-12}$ m s$^{-2}$ (dashed line) at $r_t \approx 12.6$ kpc for $M = 2.3 \times 10^{10}\, M_\odot$. For $r < r_t$ (shaded, Chameleon regime), screening dominates; for $r \geq r_t$ (blue, k-mouflage/AQUAL regime), the kinetic branch governs.

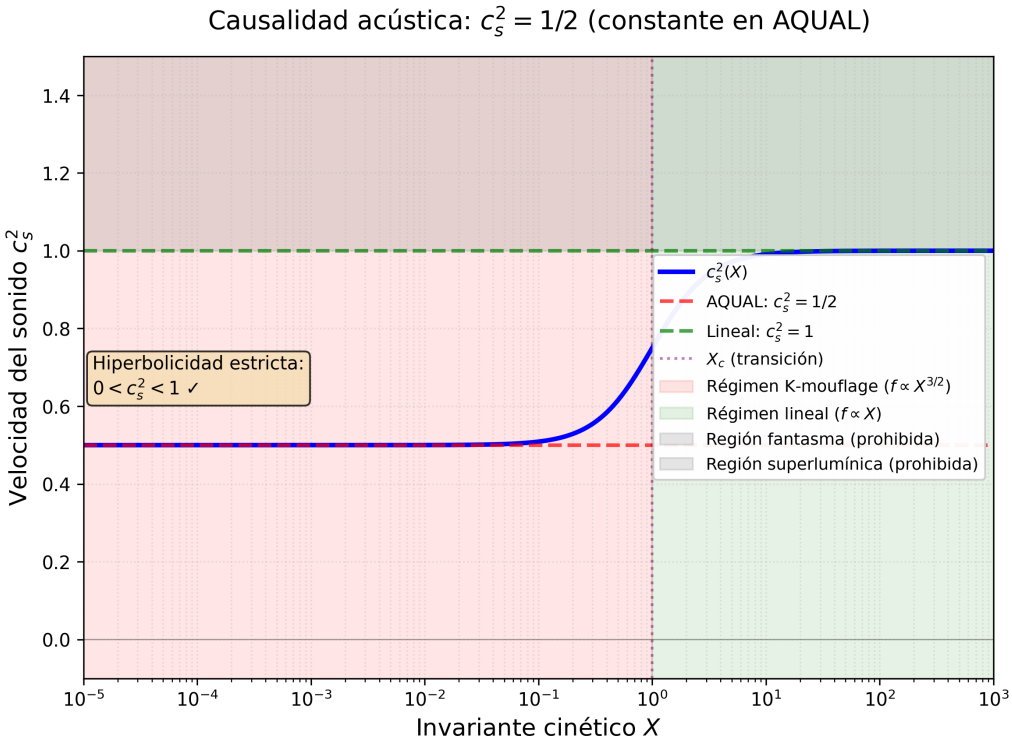


Figure 4: Effective sound speed $c_s^2$ as a function of the kinetic invariant $X$. In the AQUAL/k-mouflage regime ($X \ll X_c$, relevant to galactic scales), $c_s^2 = 1/2$ exactly. In the Newtonian limit ($X \gg X_c$, Solar-System scales), $c_s^2 \to 1$. The permitted causal region $0 < c_s^2 < 1$ ensures strict hyperbolicity and the absence of ghost instabilities throughout.